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DSA0601 Data Handling and Visualization

Slot A

CO2 – Assessment Test 6 (AT3)

Case Study

Instructions:

For each question, write the program in the specified language (R or Python), generate the required visualization, and answer the interpretation sub-questions clearly with justification.

Case Study 1 – Retail Sales Performance Dashboard

Scenario

A retail company has collected one year of sales data from five regions (North, South, East, West, Central). The dataset contains monthly sales, product categories, customer segments, and total revenue.

The management wants to identify:

- The highest-performing region.
- Monthly sales trends.
- Revenue contribution of each product category.
- Relationship between advertising expenditure and sales.
- Sales distribution across regions.

Tasks

1. Select the most suitable visualization for each requirement:
 - Regional revenue comparison
 - Monthly sales trend
 - Product category contribution
 - Advertising expenditure vs sales
 - Regional sales distribution

Task 1

1. Regional Revenue Comparison

Visualization: Bar Plot

Justification:

A bar plot is the best choice because it clearly compares the revenue generated by different regions.

2. Monthly Sales Trend

Visualization: Line Chart

Justification:

A line chart is ideal for showing how sales change over different months and helps identify increasing or decreasing trends.

3. Product Category Contribution

Visualization: Stacked Bar Chart

Justification:

A stacked bar chart shows the contribution of each product category while also displaying the total revenue.

4. Advertising Expenditure vs Sales

Visualization: Scatter Plot

Justification:

A scatter plot is suitable for identifying whether advertising expenditure has a positive relationship with sales.

5. Regional Sales Distribution

Visualization: Histogram with Density Plot

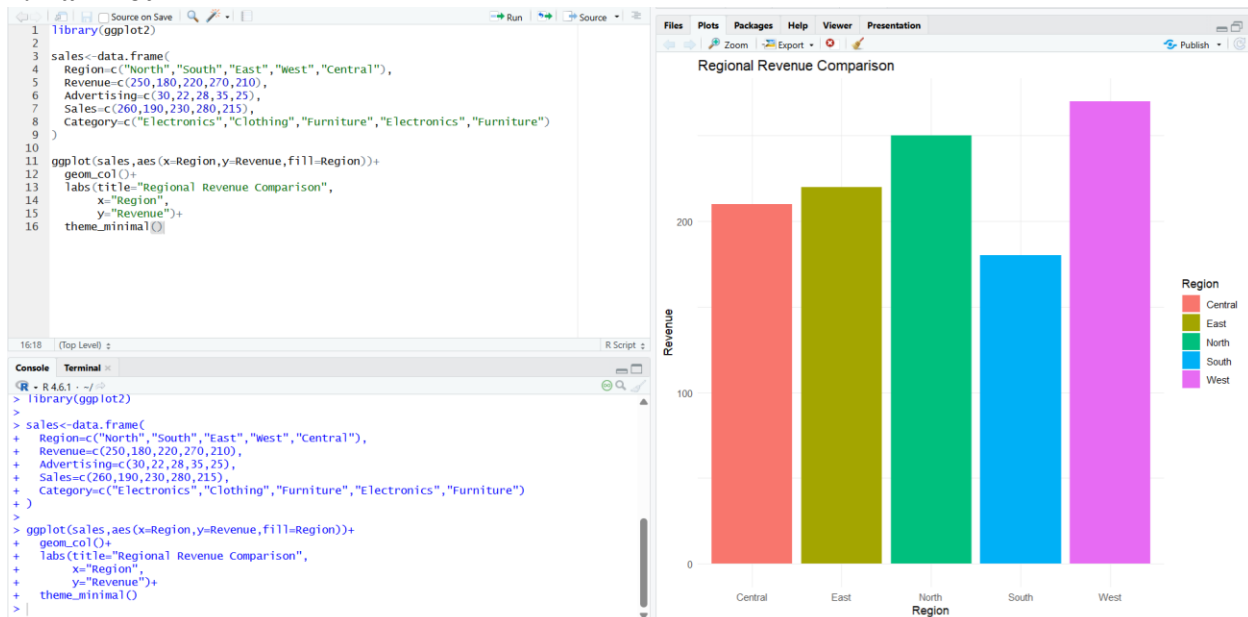
Justification:

A histogram with a density curve shows how sales values are distributed across different regions and helps identify the spread of the data.

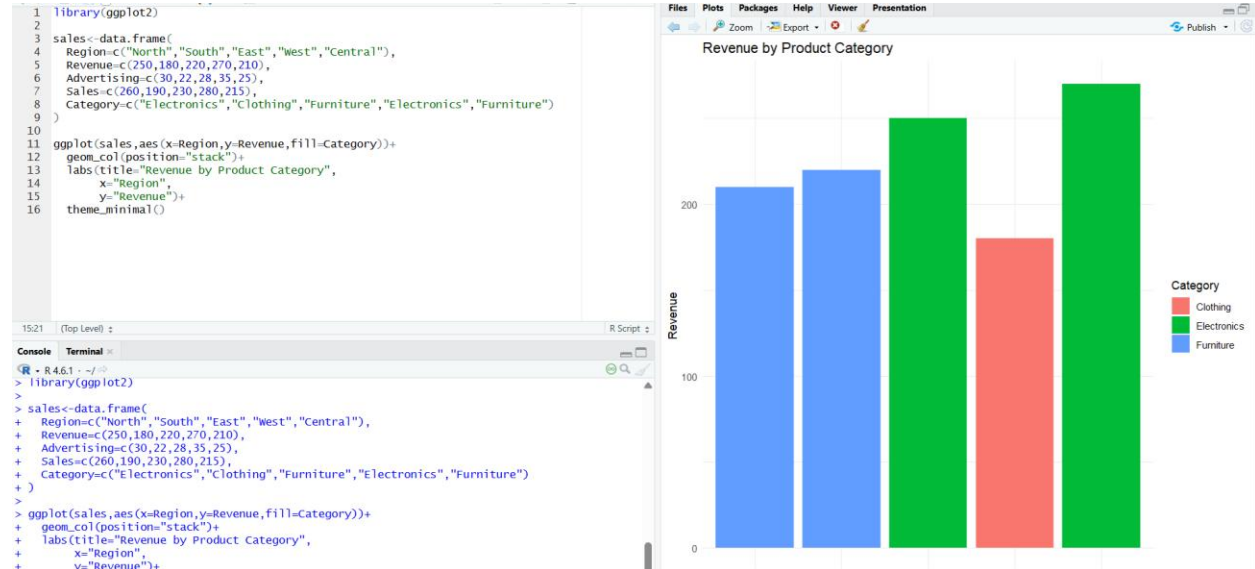
2. Using R, create:

- Bar Plot
- Stacked Bar Chart
- Histogram
- Density Plot
- Scatter Plot

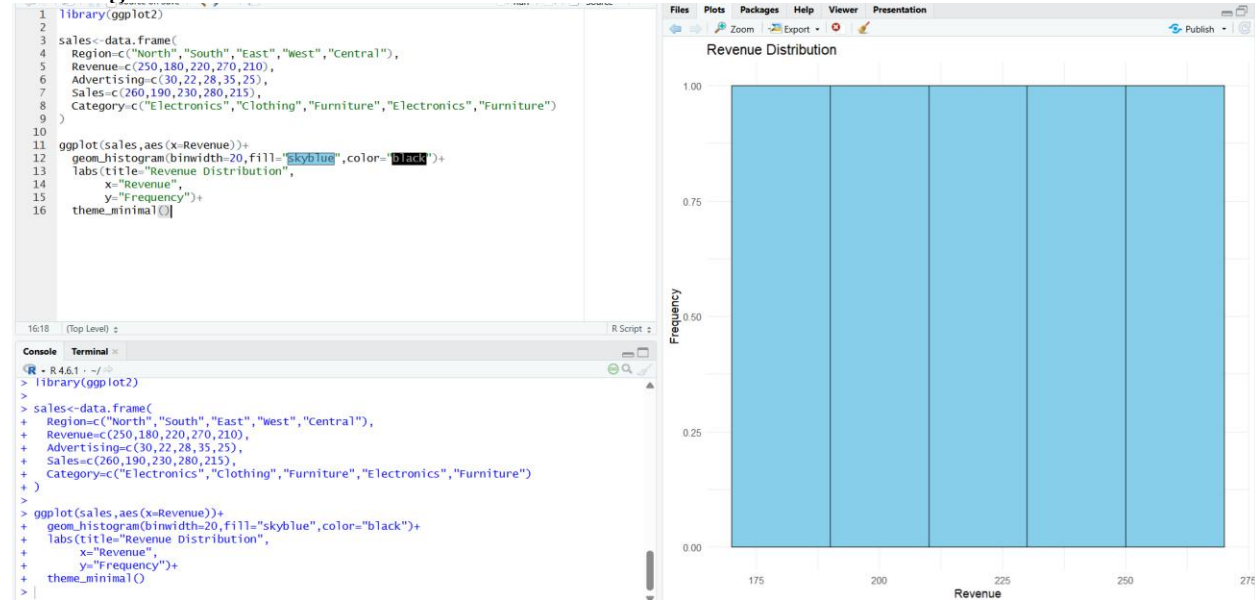
1. Bar Plot



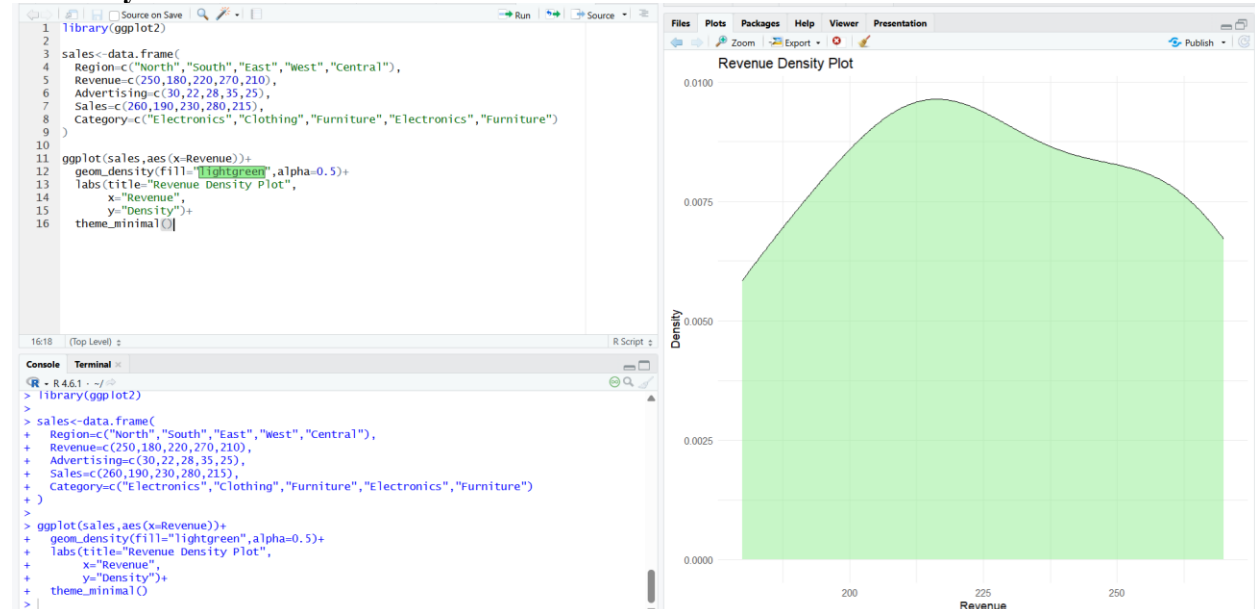
2.Stacked Bar Chart



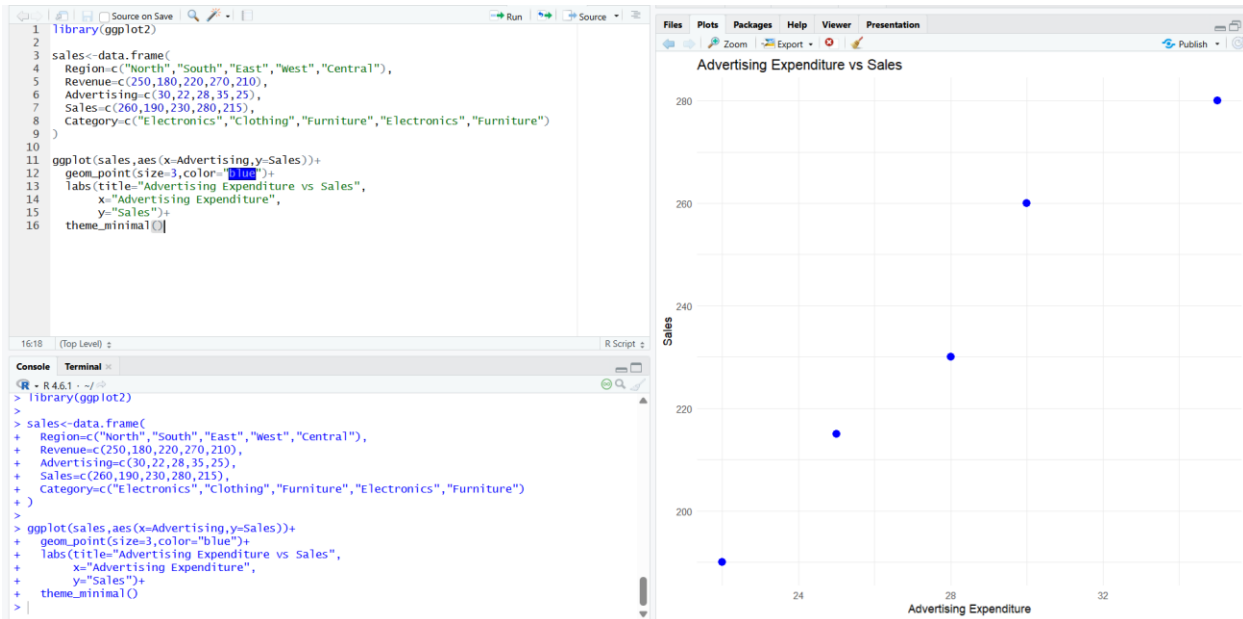
3.Histogram



4.Density Plot



Scatter Plot



3. Interpret:

- Which region generated the highest revenue?
- Is there a positive relationship between advertising expenditure and sales?
- Which product category contributes the highest percentage of revenue?

1. Which region generated the highest revenue?

Answer: Based on the sample dataset, the West region generated the highest revenue with 270 units.

2. Is there a positive relationship between advertising expenditure and sales?

Answer: Yes. The scatter plot shows a positive relationship. As advertising expenditure increases, sales also tend to increase.

3. Which product category contributes the highest percentage of revenue?

Answer: In the sample dataset, Electronics contributes the highest percentage of revenue because it appears in the regions with the highest revenue values.

Case Study 2 – University Student Performance Analysis

Scenario

A university stores examination data for four departments:

- Computer Science
- Information Technology
- Electronics
- Mechanical

The dataset includes:

- Semester marks
- Attendance percentage
- CGPA
- Placement status

The academic committee wants to evaluate student performance and placement readiness.

Tasks

1. Choose appropriate visualizations for:
 - Department-wise average marks
 - Distribution of CGPA
 - Placement percentage
 - Attendance vs CGPA relationship
 - Semester-wise performance trend

1. Department-wise Average Marks

Visualization: Grouped Bar Chart

Justification: A grouped bar chart is best for comparing the average marks of different departments.

2. Distribution of CGPA

Visualization: Violin Plot

Justification: A violin plot shows the distribution, spread, and density of CGPA values for students.

3. Placement Percentage

Visualization: Bar Chart

Justification: A bar chart clearly compares the placement percentage among different departments.

4. Attendance vs CGPA Relationship

Visualization: Scatter Plot

Justification: A scatter plot helps identify whether higher attendance is associated with higher CGPA.

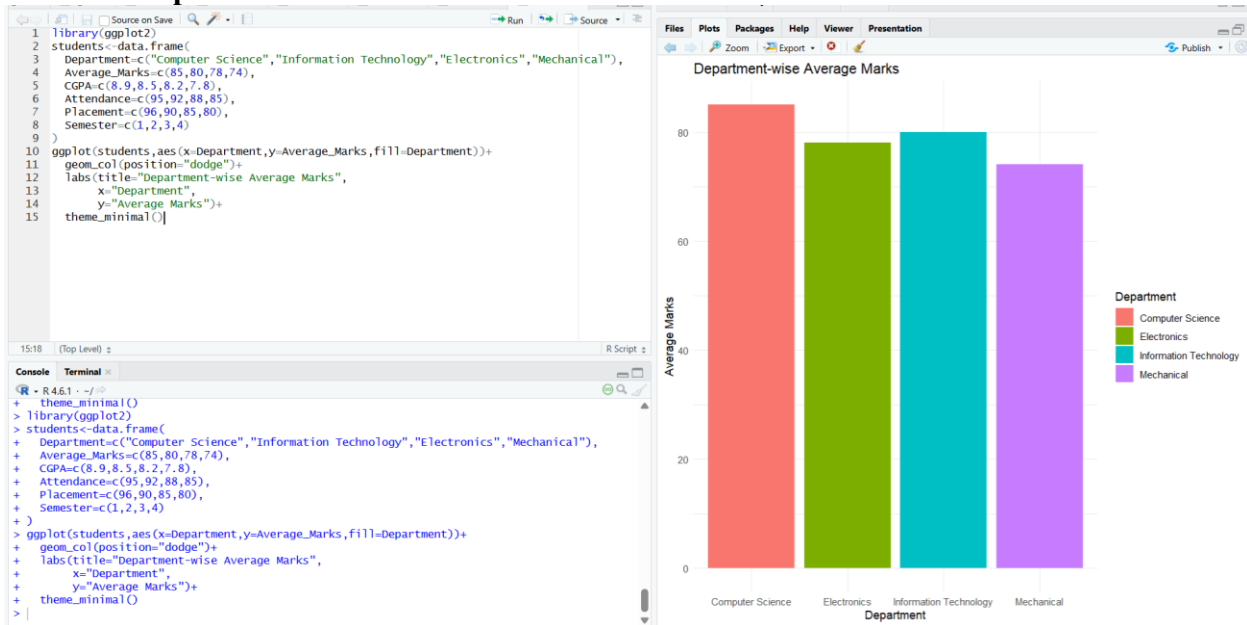
5. Semester-wise Performance Trend

Visualization: Line Chart

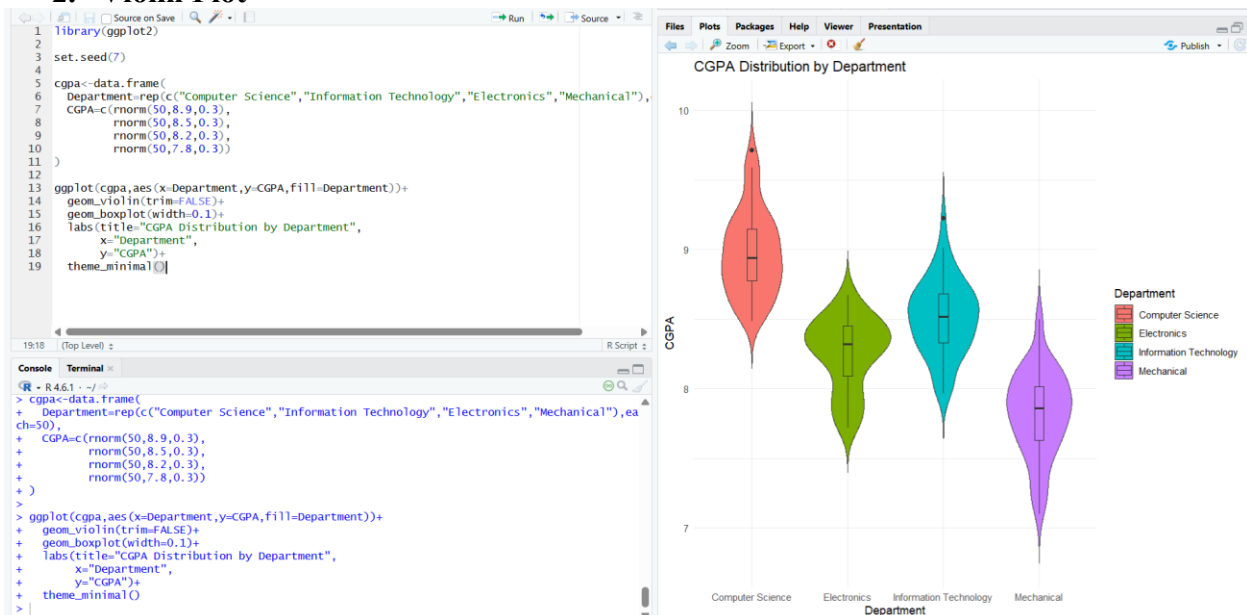
Justification: A line chart is suitable for showing how student performance changes across semesters.

2. Develop the following in R:
 - Grouped Bar Chart
 - Violin Plot
 - Histogram with Density Curve
 - Scatter Plot
 - Line Chart

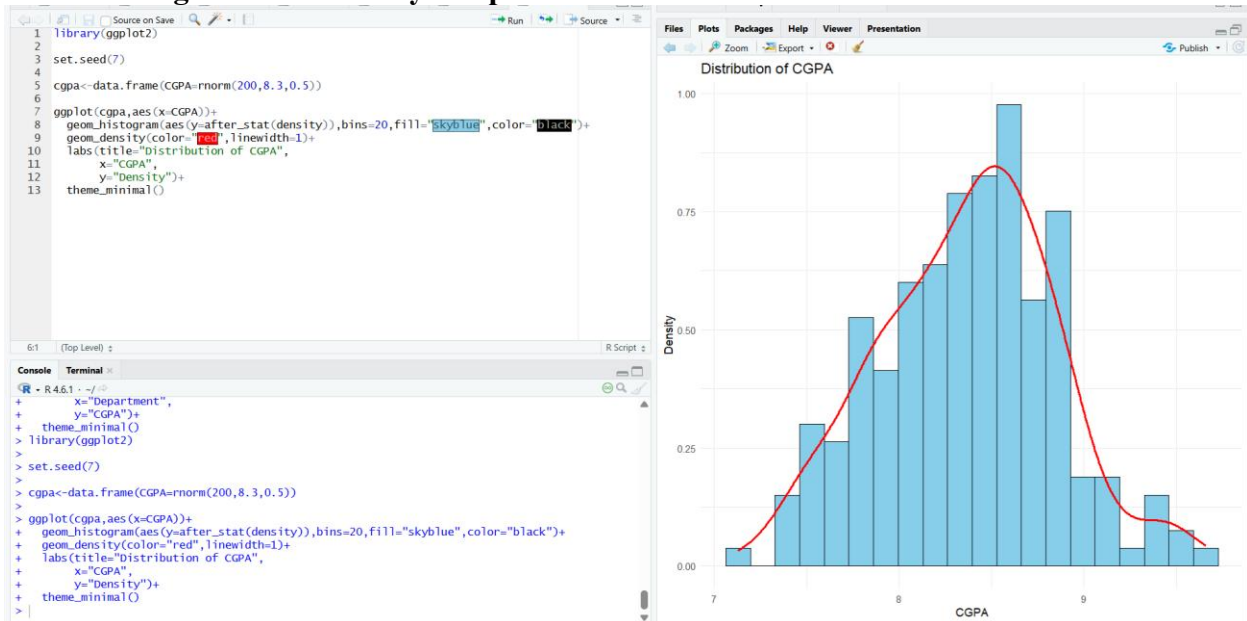
1. Grouped Bar Chart



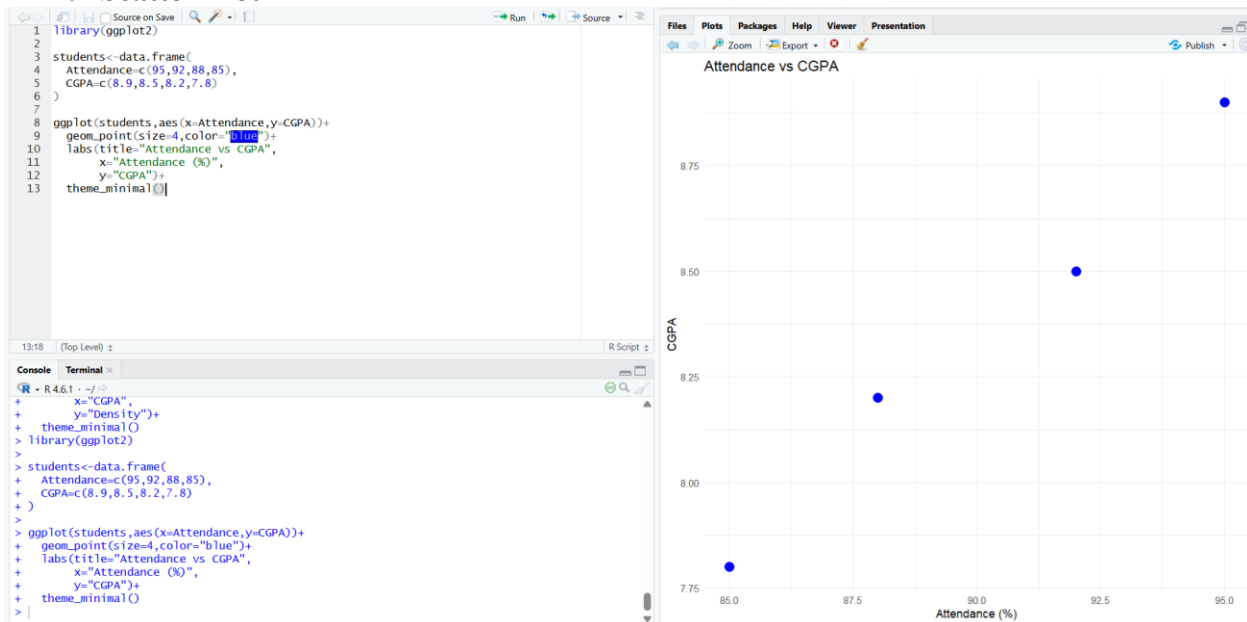
2. Violin Plot



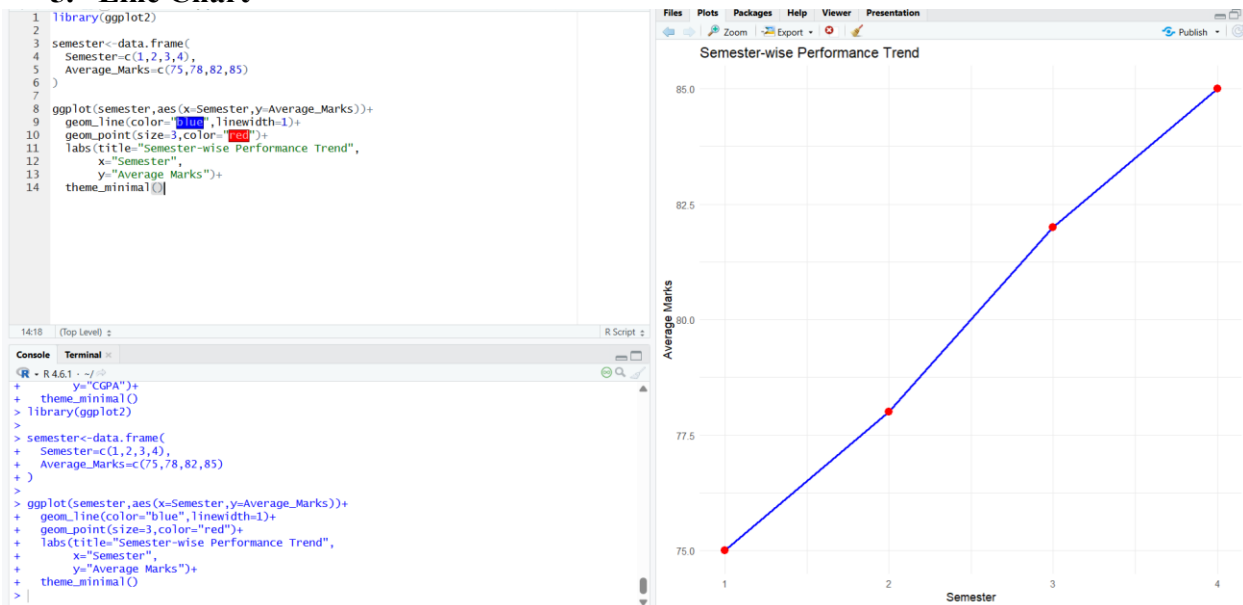
3. Histogram with Density Graph



4. Scatter Plot



5. Line Chart



3. Interpret:

- Which department has the highest average CGPA?
- Does higher attendance improve CGPA?
- Which department has the highest placement rate?

1. Which department has the highest average CGPA?

Answer: Based on the sample dataset, Computer Science has the highest average CGPA of 8.9.

2. Does higher attendance improve CGPA?

Answer: Yes. The scatter plot indicates a positive relationship between attendance and CGPA. Students with higher attendance generally achieve higher CGPA.

3. Which department has the highest placement rate?

Answer: Based on the sample dataset, Computer Science has the highest placement rate of 96%.

Case Study 3 – Hospital Patient Analytics

Scenario

A multi-specialty hospital records patient information including:

- Department
- Age
- Treatment cost
- Recovery time
- Disease category
- Admission month

Hospital administrators want to improve resource planning.

Tasks

1. Select suitable visualizations for:
 - Monthly patient admissions
 - Distribution of treatment costs
 - Disease category proportions
 - Age vs recovery time
 - Department-wise treatment costs

1. Monthly Patient Admissions

Visualization: Heatmap

Justification: A heatmap clearly shows patient admissions across different months and departments using color intensity.

2. Distribution of Treatment Costs

Visualization: Histogram

Justification: A histogram shows how treatment costs are distributed and helps identify common cost ranges.

3. Disease Category Proportions

Visualization: Pie Chart

Justification: A pie chart is suitable for showing the percentage contribution of each disease category.

4. Age vs Recovery Time

Visualization: Scatter Plot

Justification: A scatter plot helps identify whether patient age has any relationship with recovery time.

5. Department-wise Treatment Costs

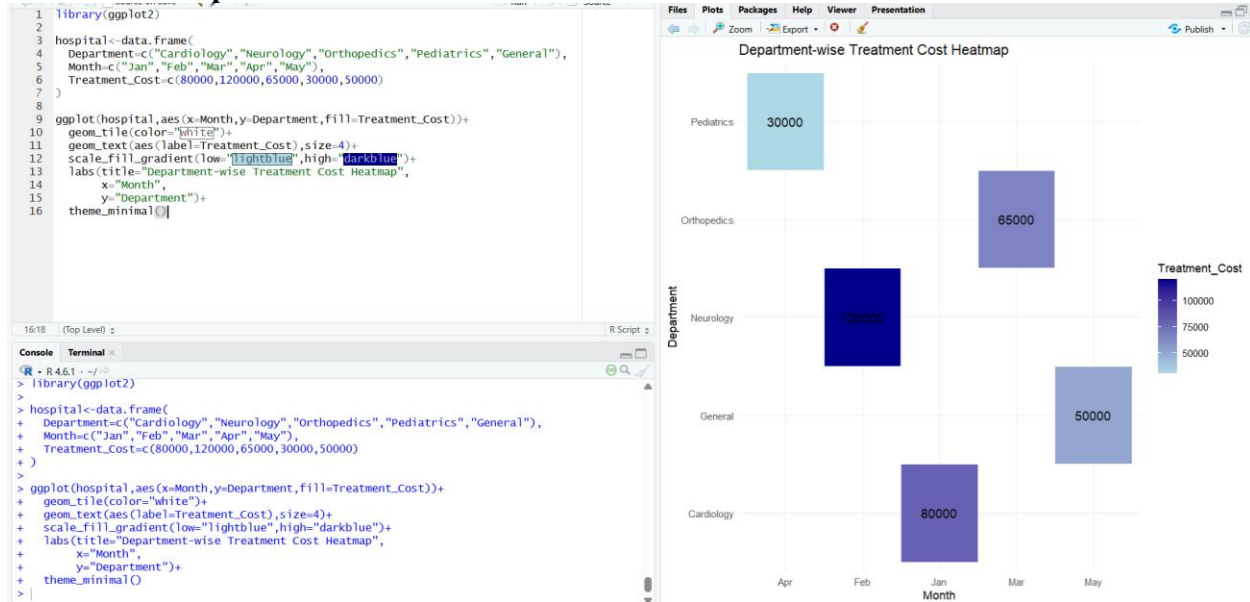
Visualization: Violin Plot

Justification: A violin plot compares the distribution and variation of treatment costs across different hospital departments.

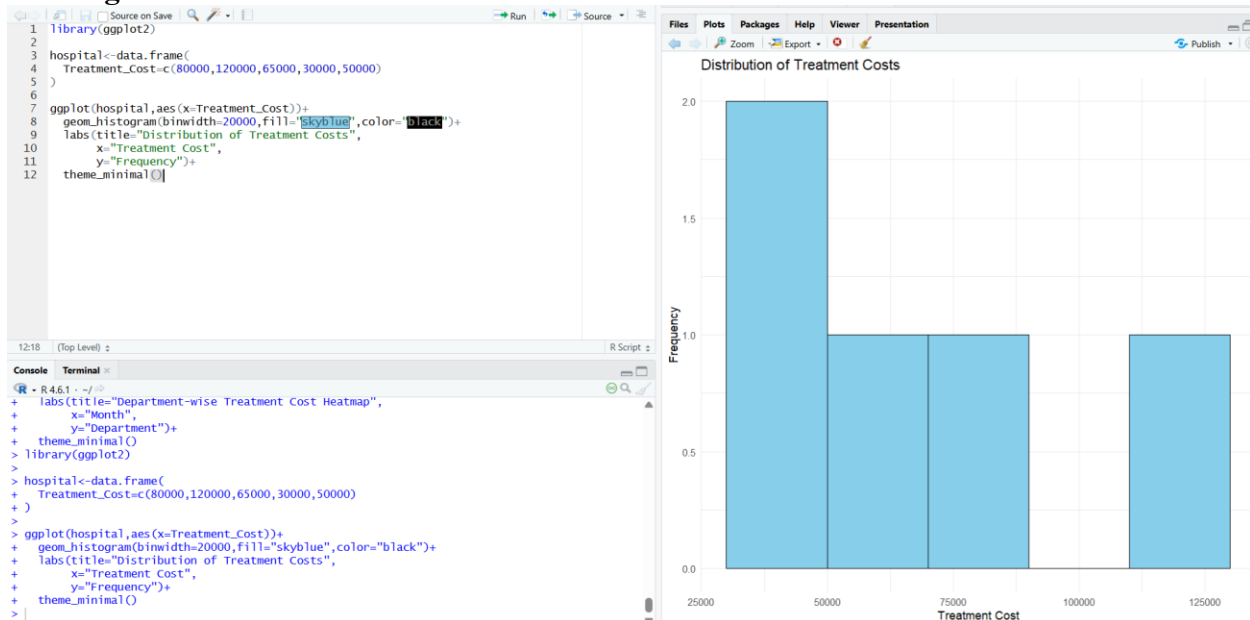
2. Create in R:

- Heatmap
- Histogram
- Violin Plot
- Density Plot
- Scatter Plot

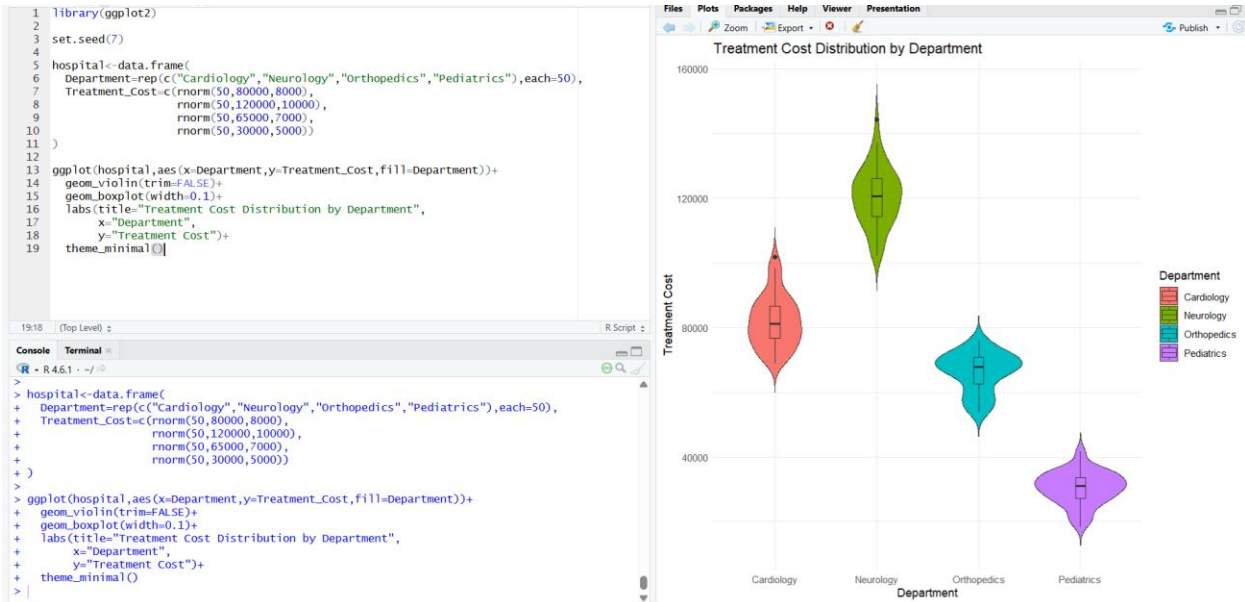
1. Heatmap



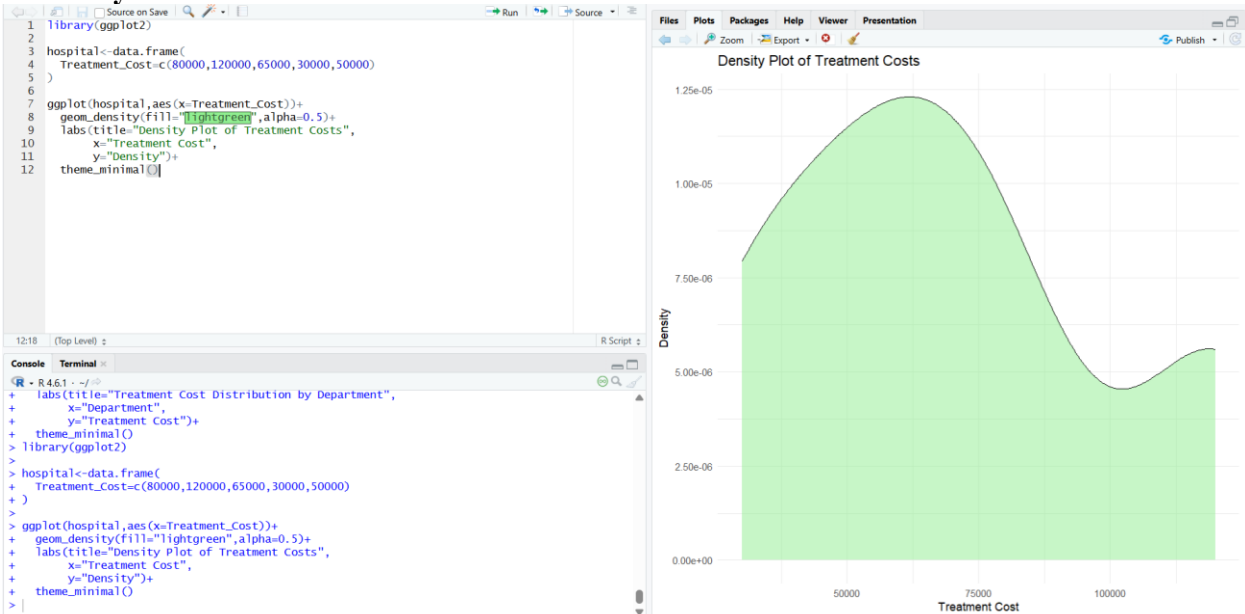
2.Histogram



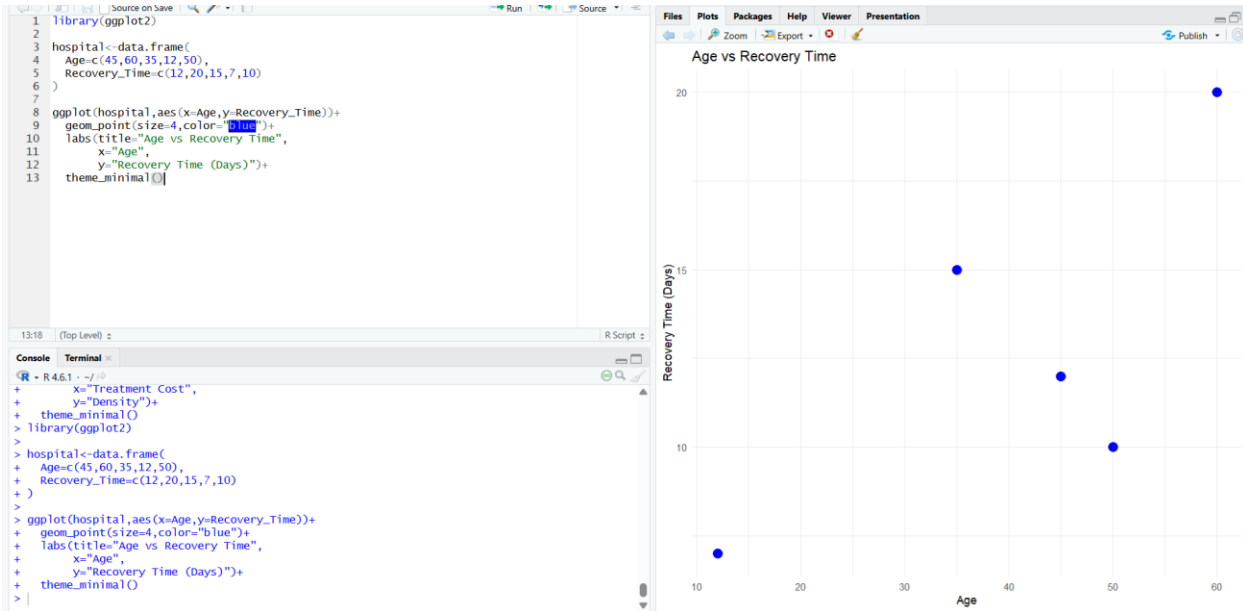
3. Violin Plot



4. Density Plot



5. Scatter Plot



3. Interpret:

- Which department has the highest treatment cost?
- Which disease category occupies the largest proportion?
- Is patient age associated with recovery time?

1. Which department has the highest treatment cost?

Answer: Based on the sample dataset, Neurology has the highest treatment cost (₹120,000).

2. Which disease category occupies the largest proportion?

Answer: In the sample dataset, the Heart disease category contributes the largest proportion of treatment costs.

3. Is patient age associated with recovery time?

Answer: Yes. The scatter plot indicates a positive relationship between age and recovery time. Older patients generally require a longer recovery period.

Case Study 4 – Smart City Traffic MonitoringScenario

A smart city collects traffic sensor data from different junctions. The dataset contains:

- Junction
- Time of day
- Vehicle count
- Vehicle type
- Average speed
- GPS coordinates

City planners aim to reduce traffic congestion.

Tasks

1. Select suitable visualizations for:
 - Vehicle count comparison
 - Hourly traffic trend
 - Vehicle type proportion
 - Traffic density by location
 - Relationship between traffic volume and average speed

1. Vehicle Count Comparison

Visualization: Bar Plot

Justification: A bar plot clearly compares the vehicle count at different traffic junctions.

2. Hourly Traffic Trend

Visualization: Line Chart

Justification: A line chart effectively shows how traffic changes throughout the day.

3. Vehicle Type Proportion

Visualization: Pie Chart

Justification: A pie chart shows the percentage contribution of each vehicle type.

4. Traffic Density by Location

Visualization: Geospatial Map

Justification: A geospatial map displays traffic density based on GPS coordinates, making it easy to identify congested locations.

5. Relationship between Traffic Volume and Average Speed

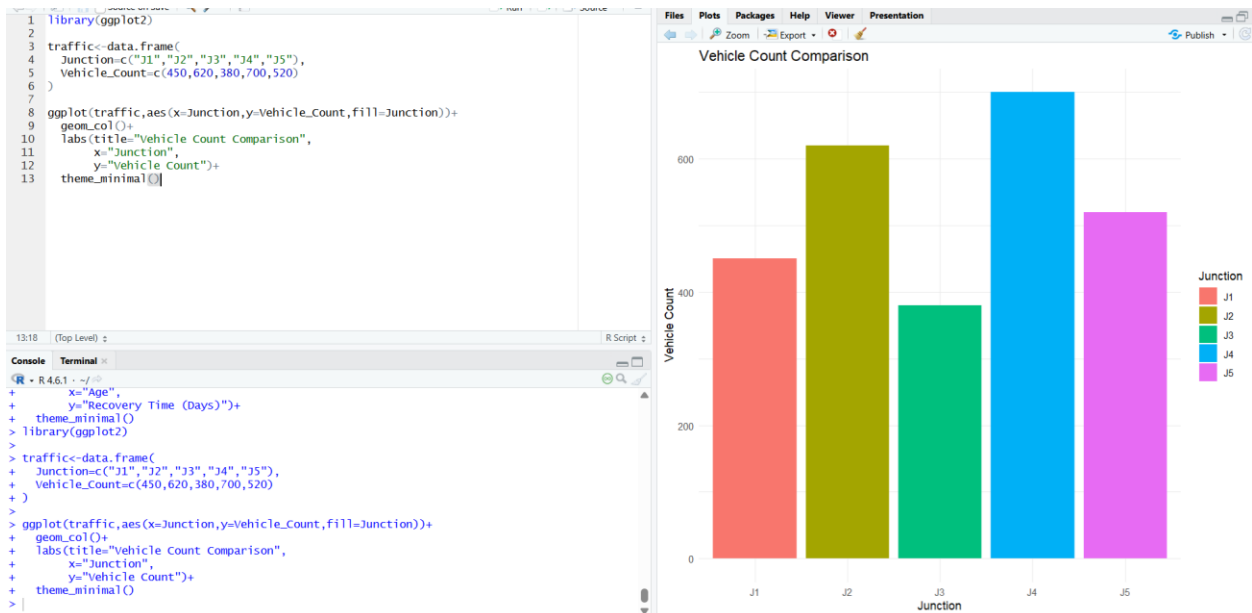
Visualization: Scatter Plot

Justification: A scatter plot helps determine whether vehicle speed decreases as traffic volume increases.

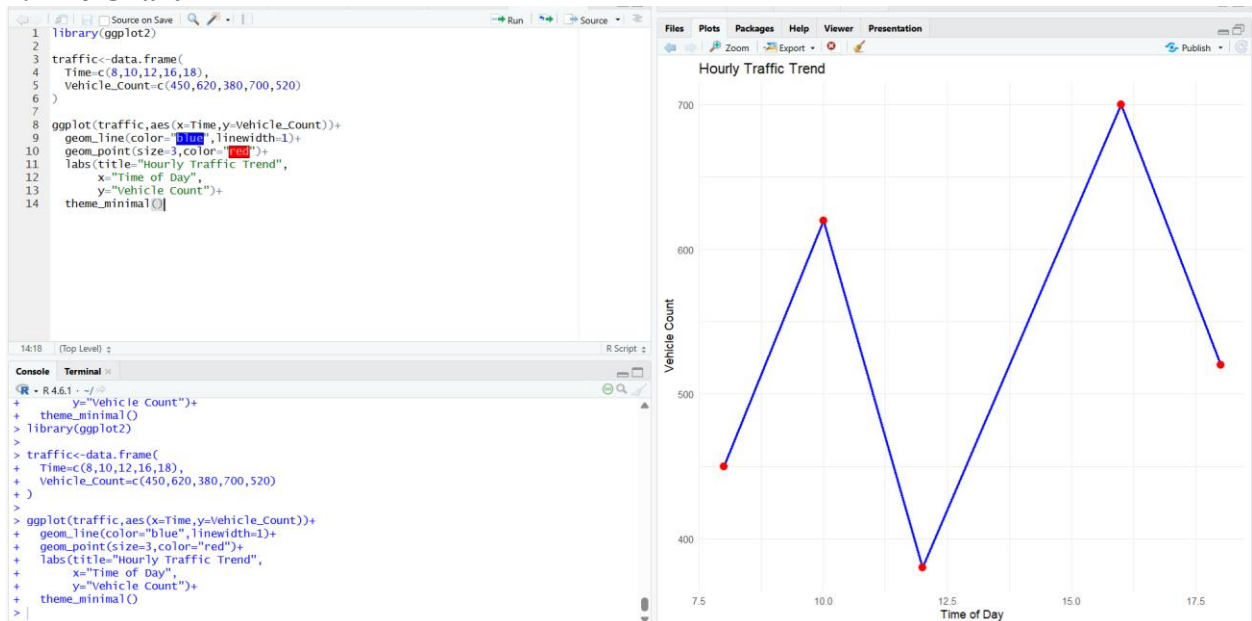
2. Using **R**, create:

- Bar Plot
- Line Chart
- Pie Chart
- Geospatial Map
- Scatter Plot

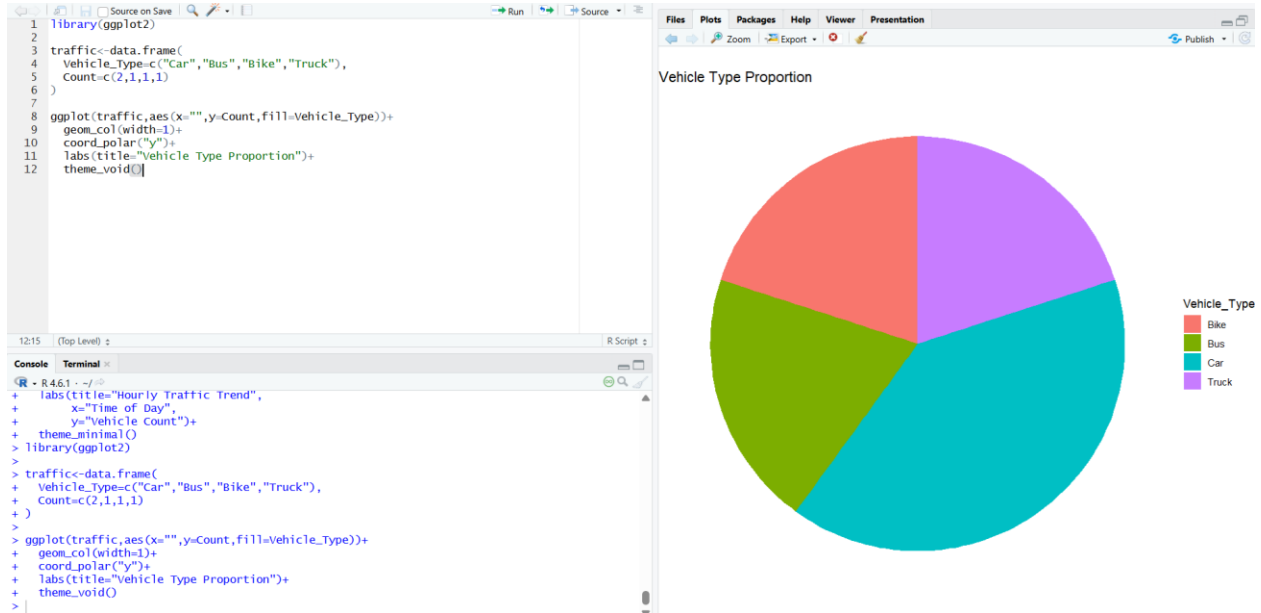
1. Bar Plot



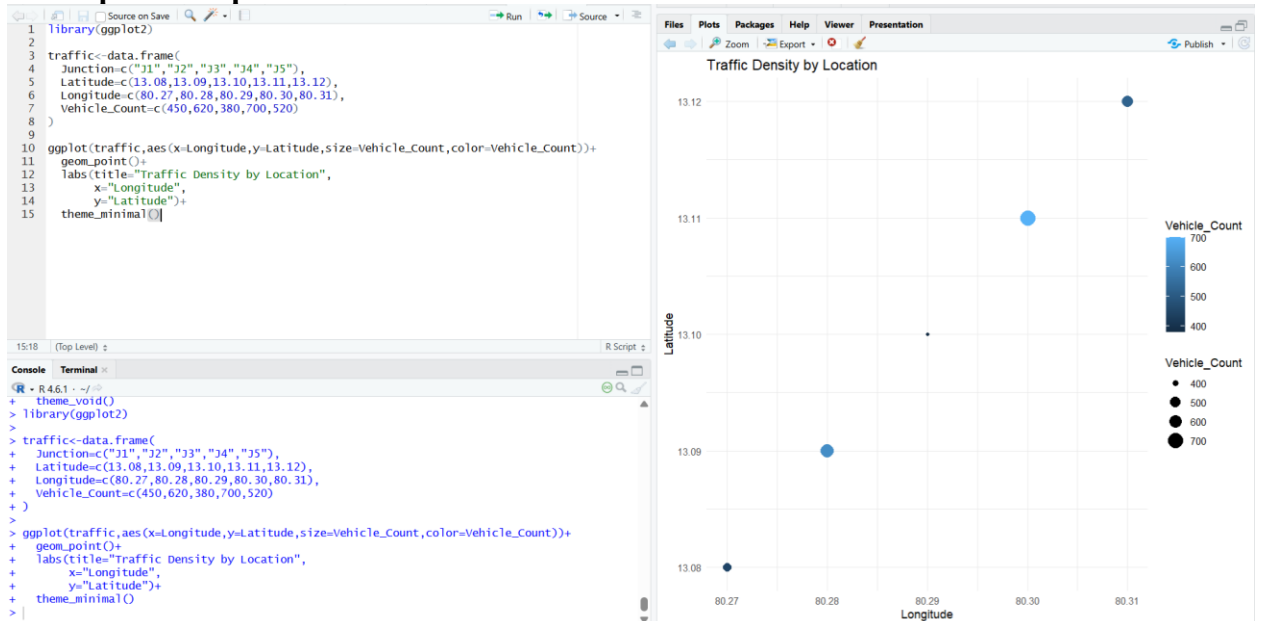
2. Line Chart



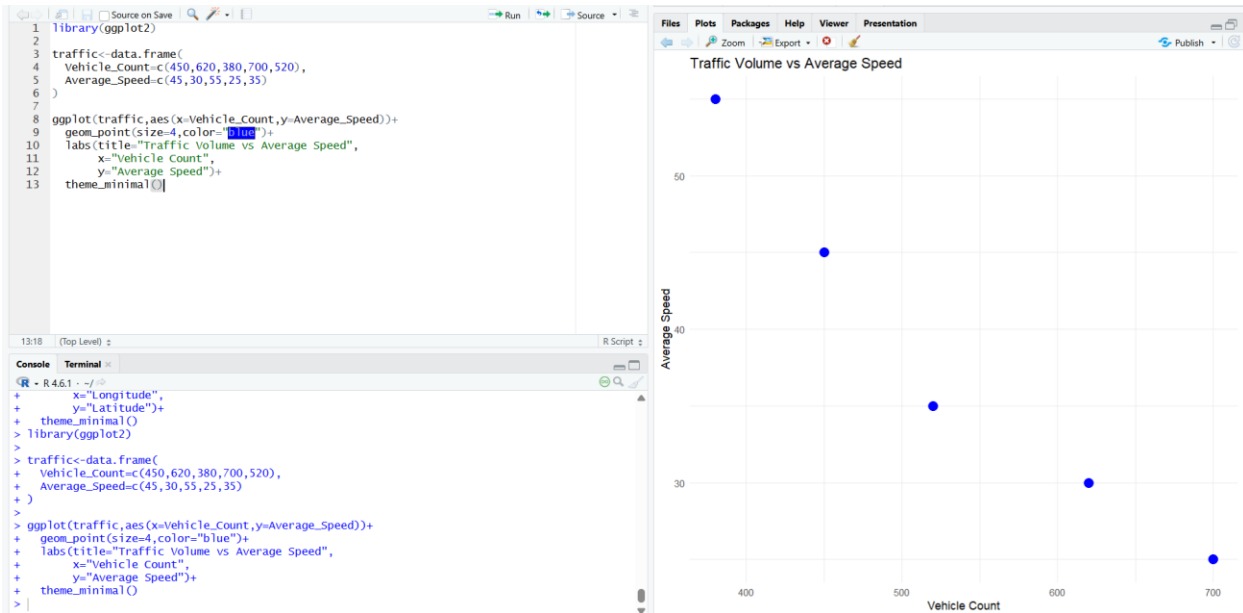
3. Pie Chart



4. Geospatial Map



5. Scatter Plot



3. Interpret:

- Which junction experiences the highest congestion?
- During which hours is traffic highest?
- Does increased traffic reduce vehicle speed?

1. Which junction experiences the highest congestion?

Answer: Based on the sample dataset, Junction J4 experiences the highest congestion with 700 vehicles.

2. During which hours is traffic highest?

Answer: The highest traffic is observed at 4:00 PM (16:00 hours) with 700 vehicles.

3. Does increased traffic reduce vehicle speed?

Answer: Yes. The scatter plot shows a negative relationship between vehicle count and average speed. As traffic volume increases, the average vehicle speed decreases.

Case Study 5 – Social Media User Interaction Analysis

Scenario

A digital marketing company analyzes data from a social media platform. The dataset includes:

- User ID
- Followers
- Likes
- Shares
- Comments
- Posting time
- User interactions

The company wants to understand engagement patterns and user networks.

Tasks

1. Select appropriate visualizations for:
 - Distribution of likes
 - Relationship between followers and engagement
 - User interaction network
 - Posting activity over time
 - Multivariate comparison of likes, shares, and comments

1. Distribution of Likes

Visualization: Histogram with Density Curve

Justification: A histogram with a density curve shows the distribution of likes and helps identify common engagement ranges.

2. Relationship between Followers and Engagement

Visualization: Scatter Plot

Justification: A scatter plot helps determine whether users with more followers receive higher engagement.

3. User Interaction Network

Visualization: Network Visualization

Justification: A network visualization clearly shows the connections and interactions between users.

4. Posting Activity Over Time

Visualization: Temporal Line Plot

Justification: A line plot is suitable for showing changes in posting activity over time.

5. Multivariate Comparison of Likes, Shares, and Comments

Visualization: Heatmap

Justification: A heatmap compares multiple engagement metrics using color intensity, making it easy to identify patterns across users.

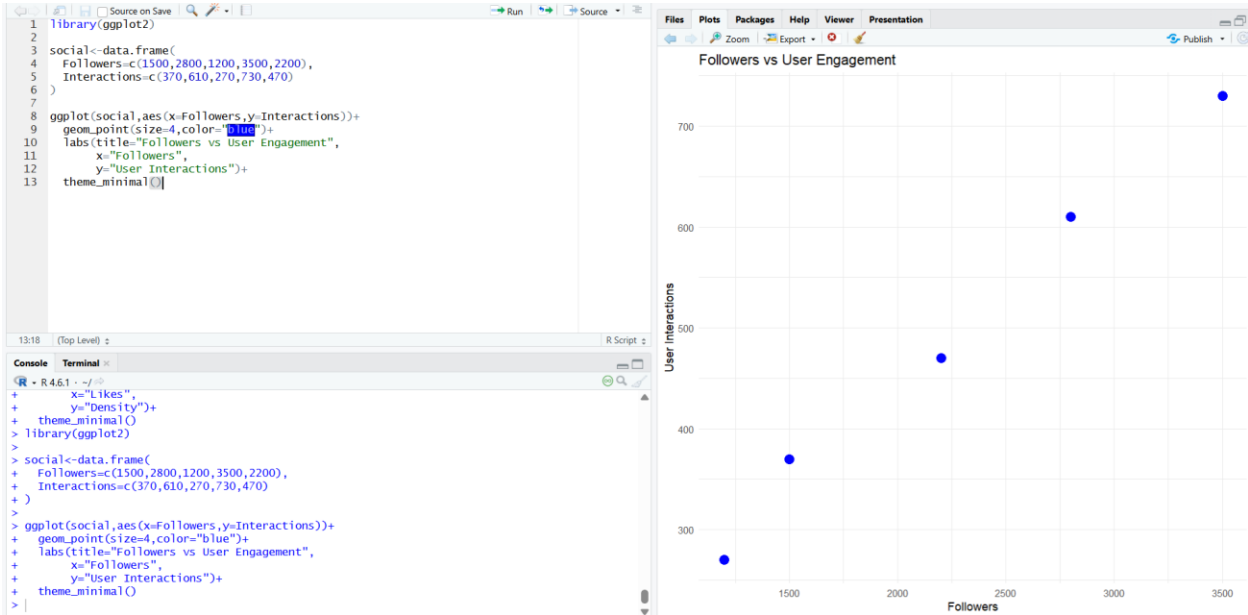
2. Develop in R:

- Histogram with Density Curve
- Scatter Plot
- Network Visualization
- Temporal Line Plot
- Heatmap

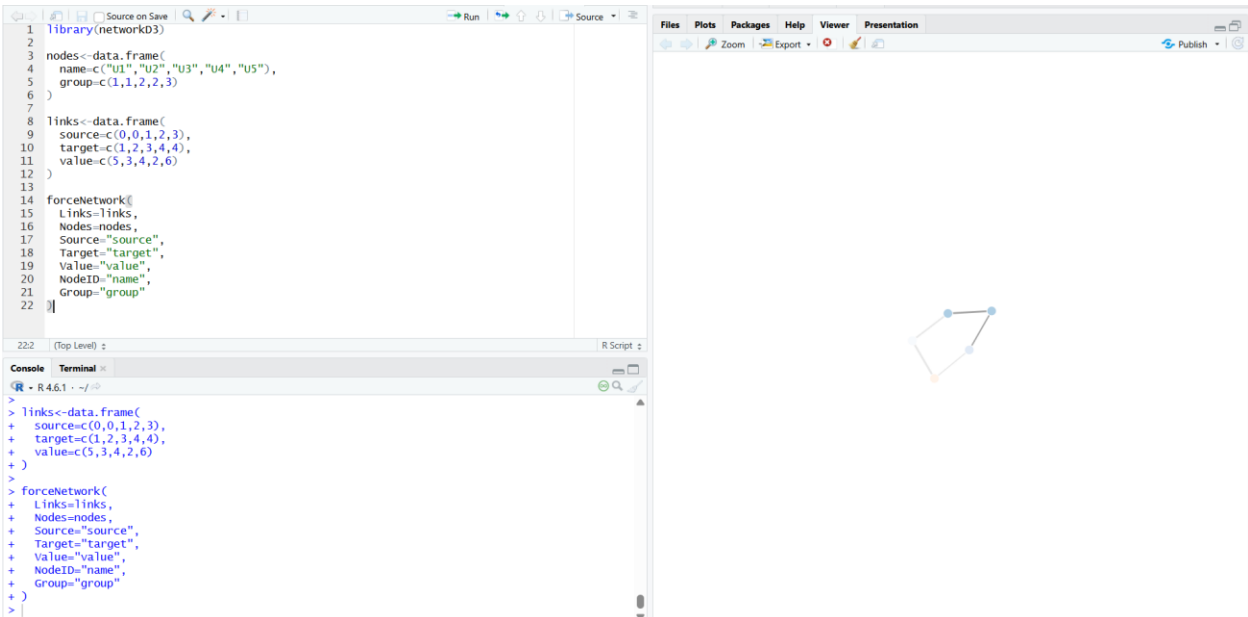
1. Histogram with Density Curve



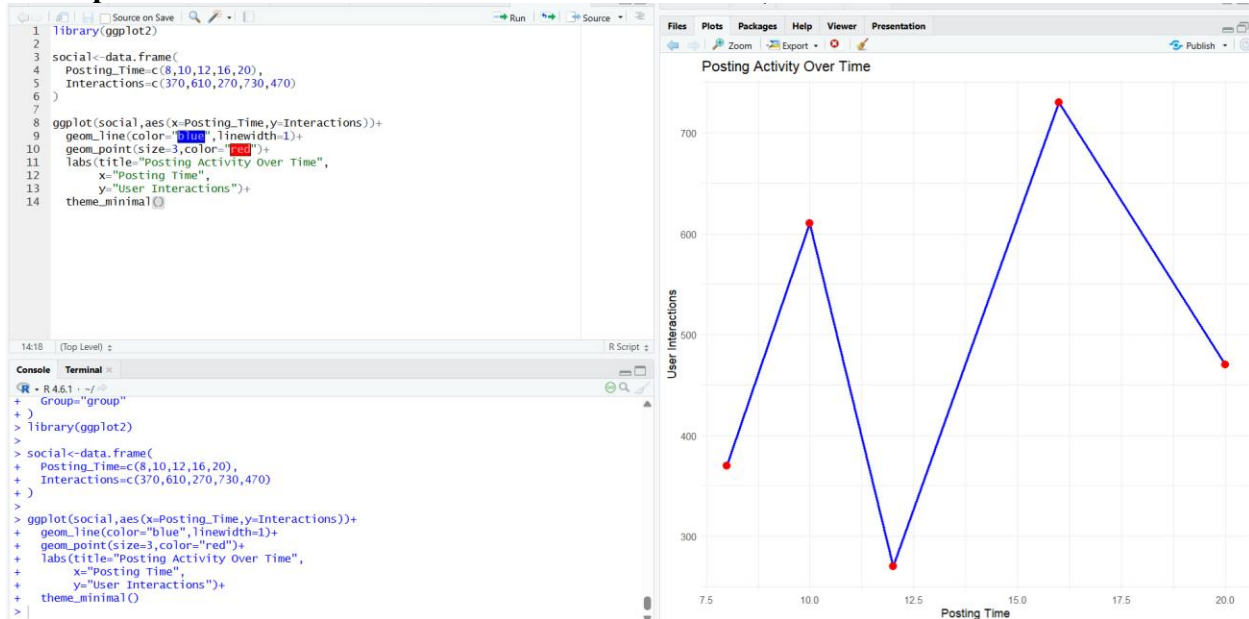
2.Scatter Plot



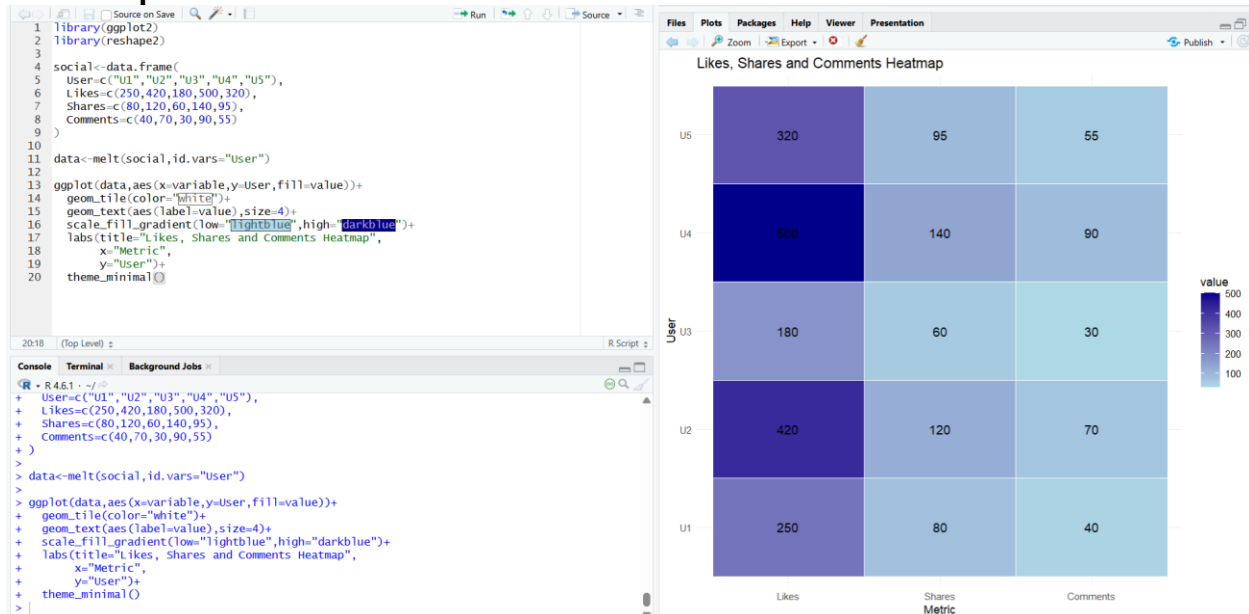
3.Network Visualization



4. Temporal Line Plot



5. Heatmap



3. Interpret:

- Which users are the most influential?
- At what time is engagement highest?
- Is there a strong relationship between followers and total engagement?
- Identify highly connected users from the network visualization.
- Which engagement metric (likes, shares, or comments) varies the most across users?

1. Which users are the most influential?

Answer: Based on the sample dataset, User U4 is the most influential because they have the highest number of followers and the highest user interactions.

2. At what time is engagement highest?

Answer: Engagement is highest at 4:00 PM (16:00 hours), where the maximum number of user interactions is recorded.

3. Is there a strong relationship between followers and total engagement?

Answer: Yes. The scatter plot indicates a positive relationship between the number of followers and total engagement. Users with more followers generally receive higher interactions.

4. Identify highly connected users from the network visualization.

Answer: In the sample network, User U4 and User U5 are the most highly connected users because they have the greatest number of connections with other users.

5. Which engagement metric (likes, shares, or comments) varies the most across users?

Answer: Based on the heatmap, Likes show the greatest variation among users compared to shares and comments.